



Mileisha Lyann Velázquez López

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RESEARCH INTERESTS

Focused on how physical, geological, chemical, and biological processes shaping coastal and marine ecosystems are impacted by climate change and unsustainable practices, with emphasis on nature-based solutions that sustain ecosystem services and biodiversity. Committed to translating aquatic science into actionable communication for communities and resource management to inform decision-making and advance environmental stewardship.

EDUCATION

- ❖ PhD, Environmental Engineering Sciences, University of Florida (UF), Gainesville, FL - *Present*
- ❖ MS, Oceanography, University of Puerto Rico (UPRM), Mayagüez, PR - 2023
- ❖ BS, Biology, UPRM - 2021
- ❖ AS, Natural Sciences, UPRU – 2019

AWARDS & HONORS

2026

- ❖ Student Travel Award – UF Water Institute
- ❖ Student Travel Award – American Geophysical Union (AGU) Annual Meeting San Francisco, CA

2025

- ❖ Student Travel Award – AGU Ocean Sciences Meeting (AGU-OSM) at Glasgow, Scotland 2026
- ❖ Graduate Student Council Travel Grant – UF
- ❖ Howard T. Odum Scholarship Award, UF
- ❖ Roy W. Linkins Scholarship, Florida Section American Water Works Association (FSAWWA)

2024

- ❖ William V. Storch Graduate Award, American Water Resources Association (AWRA)
- ❖ Sally and William Glick Graduate Research Endowment, UF
- ❖ Global Change Research Fellows Program, Southeast Climate Adaptation Science Center (SECASC) 2024 – 2025

2023

- ❖ Herbert Wertheim College of Engineering Dean's Research Award, UF (2023-2027)
- ❖ Kirkland Fellowship, UF

RESEARCH EXPERIENCE

- ❖ Research Assistant, USGS/UF, Ecological Impacts of Per- and Polyfluoroalkyl Substance (PFAS) Mixtures in a Florida Estuary, 2023 - *Present*
- ❖ Global Change Research Fellow, Southeast Climate Adaptation Science Center (SE CASC), 2024-2025
- ❖ Research Assistant, NOAA/UPRM, Ecology and Fisheries Biology of the Yellowtail snapper (*Ocyurus chrysurus*) in Puerto Rico, 2022 - 2023
- ❖ Summer Research Intern, U.S Army Corps of Engineers Engineering Research Development Center, Military Engineering Technology for In-Field Waste, 2022

WORK EXPERIENCE

- ❖ Teaching Assistant Cellular Physiology Laboratory, UPRM, 2021 - 2022
- ❖ Research Project, UPRM, Jobos Bay National Estuarine Research Reserve (JBNERR) Water Quality and Mangrove Deforestation Monitoring and Remote Sensing, 2022
- ❖ Middle School Mathematics Tutor at “UPR Estudia Contigo” Program, 2021

SKILLS

- ❖ Software & Data Tools: ArcGIS, QGIS, SentinelHub, Planet, Giovanni (NASA Earthdata), CVAT image annotation software; proficient in Microsoft Office, Google Suite, Windows/Mac OS.
- ❖ Programming & Analysis: R, basic C++; PFAS LCMS/MS analysis; Xcalibur software; calorimetry studies.
- ❖ Field & Lab Techniques: Environmental sampling (surface water, sediment, SAV, fish); microcosm experiments; DNA/RNA extraction, Bioinformatics, qPCR, chromatography, electrophoresis, microscopy, pipetting.
- ❖ Fisheries & Ecosystem Assessment: Fisheries data collection and analysis; characterization of aquatic systems.
- ❖ Scientific Communication: Proposal writing/redaction, technical writing, workshop facilitation.

- ❖ Professional Skills: Adaptable, team-oriented, strong leadership; experienced in multidisciplinary research environments.

ORAL PRESENTATIONS

- ❖ AGU-OSM Annual Meeting; Glasgow, Scotland - 2026
- ❖ AIChE Annual Meeting; San Diego, CA - 2024
- ❖ UF Water Institute Symposium - 2024
- ❖ UF Water Institute Symposium – 2024

POSTER PRESENTATIONS

- ❖ AGU Annual Meeting San Francisco, CA - 2026
- ❖ AEESP Conference – 2023 & 2024
- ❖ Graduate Student Research Day, UF - 2024
- ❖ Engineering Research Symposium, UF - 2024
- ❖ Graduate Student Research Day, UF - 2024
- ❖ Engineering Research Symposium, UF - 2024
- ❖ OSRC Summit, UF - 2023

LANGUAGES

- ❖ Bilingual in Spanish and English, Full professional proficiency

LICENSES

- ❖ Open Water Scuba Diving & Boating license

RELEVANT COURSEWORK

- ❖ *Environmental Genomics/Bioinformatics*
 - Explore applications of environmental DNA for ecosystem monitoring, management and water quality assessments.
 - Analytical skills acquired include analysis of metabarcoding data, mapping reference genomes, and quality control analysis.
 - Explored the 'Bismark' and 'methylKit' analysis pipeline to process whole-genome bisulfite sequencing (WGBS) data.
- ❖ *Ecosystem Engineers*
 - Understand the roles of ecosystem engineers and learn the traits that define them through discussion of scientific literature.
 - Acquire insights on ecosystem engineering research perspectives for application to relevant research interests.
- ❖ *Water, Environment, and Human Health*

- Implementation of computer analysis applications for engineering of water, environment, and human health issues.
- Explored how sociological perspectives inform approaches to environmental complexity and justice
- Integrate analytical and technological approaches for the improvement of community health, including water and climate systems.

❖ *Marine Pollution*

- Sources of marine pollution
- Develop critical thinking skills by studying case studies in Puerto Rico and their potential solutions.
- Use mathematical skills to estimate fluxes of pollutants in the oceanic reservoirs.

❖ *Biological Oceanography*

- Study the biological processes that occur within the distinct marine ecosystems and how they are influenced by external factors. These general factors or parameters include human activity which contribute to climate change, coastal erosion, and pollution.
- Identify and directly monitor changes to coastal features and benthic habitats in the Natural Reserve of La Parguera, Lajas, PR.

❖ *Chemical Oceanography*

- Chemical composition of seawater and vital chemical reactions within the oceans
- Ocean acidification and harmful toxins that pollute marine ecosystems
- Final project on Neon, a rare element found in our oceans, and identifying its presence, speciation, depth profile, fluxes, and residence time within our oceans.

❖ *Marine Geology*

- Geological processes that occur within our coasts and seafloor, influence continental margins and affect stability of coastal infrastructures.
- Develop a topic involving marine geologic processes, considering subduction zones at trench areas and the geological controls that influence trench fauna.

❖ *Remote Sensing*

- Successfully develop and present a project based on monitoring progression of mangrove deforestation at a construction site in an area within the Jobos Bay National Estuarine Research Reserve (JBNERR)
- Analyze satellite data using QGIS and ArcGIS programs
- Interpret water quality parameters which in hand explain environmental processes.
- Satellite and in situ data analysis of coastal water quality parameters before and after Fiona storm in Puerto Rico
- Recommend future island coastal development based on ecological engineering practices to reduce pollution and biodiversity loss.

❖ ***Soil Microbial Ecology***

- Analyze physiological foundations for microbial activities in soil and apply them to ecological interactions and processes.

- Discuss the fundamental physiology and ecology underlying essential biogeochemical cycles in carbon, nitrogen, and sulfur cycles.
- Identify appropriate methods for addressing fundamental questions in soil microbial ecology.

❖ ***Independent Study: Science Communication of the Southeast Workshop (SECASC Fellowship)***

- Engage in knowledge translation by summarizing workshop sessions concerning drought in southeastern aquatic ecosystems with federal, Tribal, regional, and state natural resource managers, water resource managers, drought planners, regional boundary organizations (water and climate), scientists, and data providers.
- Organizing tools and resources to ensure implementation beyond workshop discussions.
- Facilitating discussions on identifying barriers and opportunities to improve preparedness for future droughts.